

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No.....

EX-501 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Utilization of Electrical Energy
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. A 20 kW; Single phase, 220 V resistance over employs circular nichrome wire for its heating element, if the wire temperature is not to exceed 1227° and the temperature of the change is to be 427°C . Calculate the size and length of the wire required. Assume emissivity = 0.9, radiating efficiency = 0.6 and specific resistance of the wire = $1.09 \times 10^{-6} \Omega\text{-m}$.
2. Describe the principle of working and applications of different methods of induction heating.
3. An illumination of 25 lux is to be produced on the floor of a room $12 \text{ m} \times 9 \text{ m}$. 18 lamps are required to produce this illumination in the room, if 50% of the emitted light falls on the floor. What is the power of the lamp in Candela?
4. What is flood lighting? Explain the projector used and also describe factor used in flood lighting calculation.
5. Define the following
 - i) Plane angle ii) Solid angle
 - iii) Luminous intensity iv) Candela
6. An electric train has an average speed of 42 kmph on a level track between stops 1400 m apart. It is accelerated at 1.7 kmphps and is braked at 3.3 kmphps. Draw the speed time curve for the run.
OR
Explain Faraday's laws of electro-deposition. Also explain in brief the need of electro-deposition.
7. a) Describe the main features of
 - i) Plugging ii) Regenerative brakingb) Discuss the following
 - i) Electric drives ii) Load equalization
8. Explain the following : (Any two)
 - a) Designing of heating element b) Arc furnace transformer
 - c) Electron beam welding d) Ideal Traction system
